GG-S-598C
April 6, 1976
SUPERSEDING
GG-S-00598B(DSA-DM)
August 15, 1967 and
GG-S-598A
February 2, 1960

FEDERAL SPECIFICATION

SPATULA, DENTAL AND CARVER, DENTAL (WAX)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers double-ended spatulas and a carver suitable for dental use.

1.2 Classification.

1.2.1 Types and styles. Spatulas and carver shall be of the following types and styles, as specified (see 6.1):

Type I - Spatulas
  Style 1 - Beale, No. 7
  Style 2 - Gritman, No. 31

Type II - Carver, Roach

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

FSC 6520

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Federal Specifications:

GG-I-526 - Instruments, Dental and Surgical; General Specifications.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Setup.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment, (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

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Military Specifications:

MIL-P-116 - Methods of Preservation.
MIL-L-10547 - Liners, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

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(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society For Testing and Materials (ASTM):

E 18	-	Tests for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.
E 92	-	Tests for Vickers Hardness of Metallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 General requirements. The spatulas and the carver shall be in accordance with the applicable requirements of GG-I-526, with the exceptions and with the additional requirements specified herein.

3.1.1 Material. Spatulas and carvers shall conform to the requirements of class 4, type 420 or type 440A stainless steel.

3.1.2 Standard sample. The finish and workmanship shall be equal to or better than the standard sample (see 3.5, 3.7 and 6.2).

3.2 Hardness. After proper stress relieving, the Rockwell hardness of the operative ends shall be C40 to C50 for instruments fabricated from type 420 steel, and C52 to C56 for instruments fabricated from type 440a steel. Handles and shanks shall be properly tempered and shall have a hardness suitable for the purpose intended. Hardness shall be determined as specified in 4.4.1.

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3.3 Style, design and dimensions. The spatulas and carvers shall be double-ended instruments of one-piece construction. The instruments shall conform to the design and dimensions for the type and style as specified. The periphery of the blades of all operative ends shall be blended or tapered to form a carving edge, except for the spoon end of the carver. The instruments shall be suitable for use with or without heat for carving and spreading various waxes in the preparation of dental prosthetic appliances.

3.3.1 Type I, spatulas.

3.3.1.1 Style 1. Spatula shall be a No. 7 spatula of the Beale design, as shown in figure 2.

3.3.1.2 Style 2. Spatula shall be a No. 31 spatula of the Gritman design, as shown in figure 3.

3.3.2 Type II, carver. Carver shall be of the Roach design, as shown in figure 1.

3.3.3 Figures. The figures forming a part of this specification are specification control figures. The supplier shall be responsible for preparing his own shop drawings. Where tolerances prescribed could cumulatively result in improper design, the supplier shall provide a product with dimensions within the tolerances prescribed on the drawings to insure proper design, function, fit, assembly and operation of the item. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer.

3.4 Handles. Spatula handles shall be smooth, or shall have fillet or notch type serrations. Carver handles shall have fillet or notch type serrations. Depth of the serrations shall be 0.001 to 0.004 inch. There shall be 40 ± 10 serrations per inch on seven faces and one smooth face for identification marking.

3.5 Finish. Smooth spatula handles shall have a satin, or mirror, or bright polished finish. Those instrument handles with fillet type or notch type serrations shall have a bright finish. The operative ends and the shanks of both spatulas and the carver shall have a mirror or a bright polished finish, except for the carver, the spoon concavity may have fine hone markings. The finish of the operative ends shall be equal to or better than the standard sample (see 3.1.2 and 6.2).

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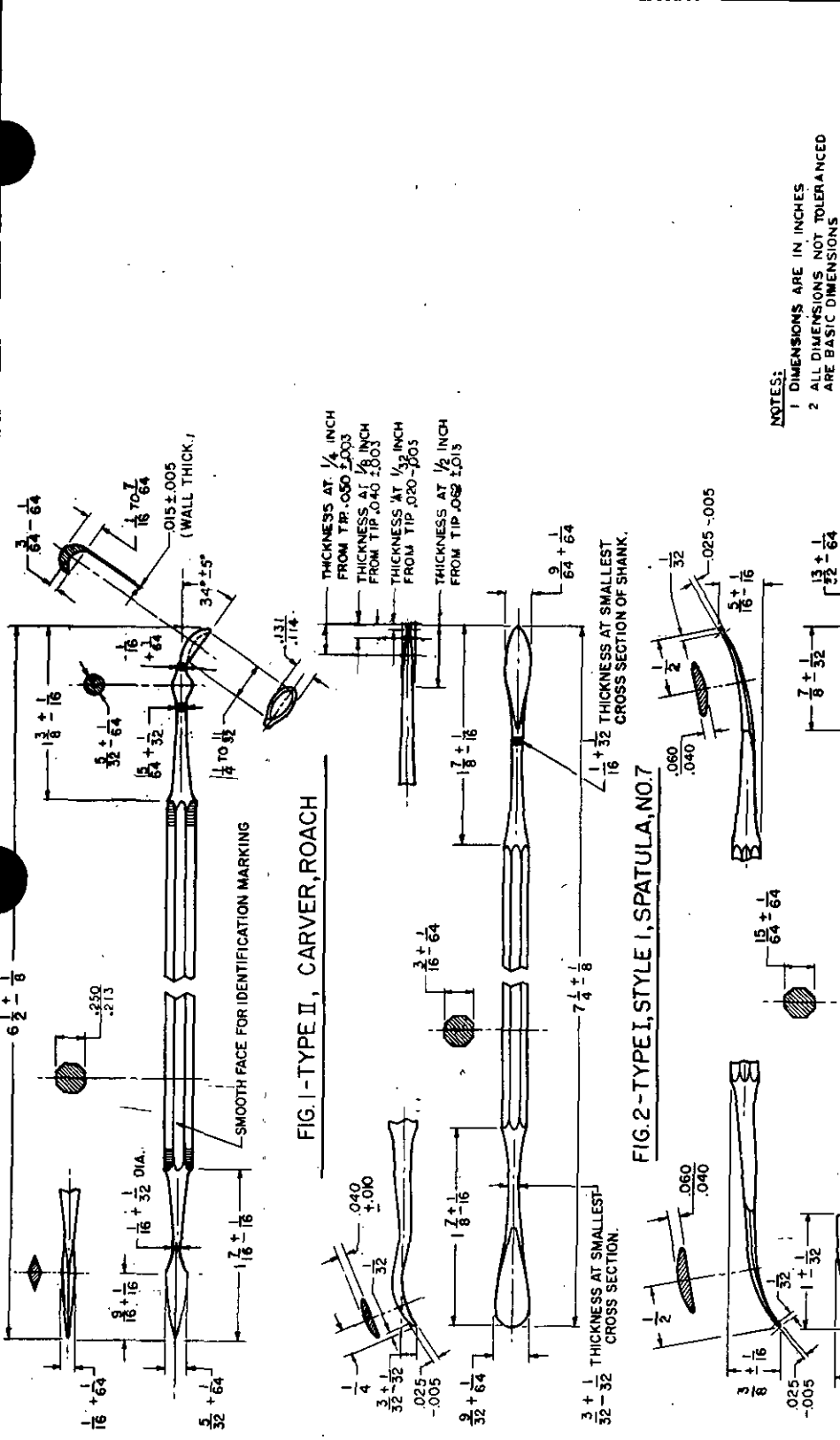


FIG. 1 - TYPE II, CARVER, ROACH

FIG. 2 - TYPE I, STYLE 1, SPATULA, NO. 7

FIG. 3 - TYPE I, STYLE 2, SPATULA, NO. 31

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3.6 Identification marking. In addition to the marking requirements of GG-I-526, each spatula shall be marked with the instrument number, and each carver shall be marked with the letters "RWC". All marking shall be legible and permanent. The word "stainless" shall also be marked on instruments of foreign manufacture. As an alternate to the word "stainless", the use of a registered trademark identifying the steel as stainless shall be acceptable, provided prior approval is received from the cognizant contracting officer.

3.7 Workmanship. In addition to the requirements of GG-I-526, the operative ends shall be free of discolorations and coarse grind marks. Workmanship shall be equal to or better than the standard sample (see 3.1.2 and 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials may be furnished in lieu of actual performance of such testing by the contractor, provided that lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in tables I, II, III and IV (see 4.2.1, 4.2.2, 4.3 and 4.3.1) and 4.3.1.1.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQL's) shall be as indicated in table I. Unit of product for examination purposes shall be the unit of issue.

TABLE I. Sampling for examination

Type defect	Inspection Level	AQL (Percent Defective)
Major A	II	1.0
Major B	S-2	1.5
Minor A	II	2.5
Minor B	S-2	4.0

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in table II. The acceptance number shall be zero for all sample sizes. Lots for hardness testing shall be as specified in 4.2.2.1.

TABLE II. Test criteria

Characteristic	Reqmt.	Test procedure	Inspection level
Chemical composition	3.1.1 and GG-I-526	GG-I-526	S-1 1/
Hardness	3.2 and GG-I-526	4.4.1	S-2
Copper sulfate	GG-I-526	GG-I-526	S-2
Boil test	GG-I-526	GG-I-526	S-2

1/ A sufficient quantity of each lot of metal used in the construction of the spatulas or carver shall be selected for composition analysis in lieu of sample instruments, provided lot identity has been maintained and can be demonstrated to the Government.

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4.2.2.1 Lots for hardness criteria. For finished instrument (operative end) hardness testing, a lot or batch of instruments shall consist of instruments produced by one manufacturer at one plant, from the same material (same raw material heat and heat treatment) and from the same heat treatment lot or batch (instruments undergoing heat treatment at the same time, and in the same heat treatment equipment).

4.2.3 Sample unit usage. Where feasible, the same sample unit may be used for the determination of two or more test characteristics.

4.3 Examination. The spatulas and carver shall be examined to determine compliance with requirements contained in this specification. Examination shall be conducted in accordance with the following classification of defects. Examination shall include, but shall not be limited to, the defects listed in table III.

TABLE III. Classification of defects

Categories	Defects
Major A:	
101	Style and design not as specified for the applicable type and style.
102	Finish on operative ends not as specified.
103	Operative ends not free of coarse grind marks, cracks, nicks, gouges, burrs or other imperfections.
104	Operative end contains bent or hooked points.
105	No carving edge, as applicable.
Major B:	
151	Operative end dimensions not as specified.
Minor A:	
201	Serrations not as specified, or incomplete, as applicable.
202	Handle not straight.
203	Finish on handle and shank not as specified.
204	Instrument identification not present, not legible, not permanent, or improperly located.

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TABLE III. Classification of defects (cont'd)

Categories	Defects
Minor A: (cont'd)	
205	Instrument not permanently and legibly marked with name or trademark of manufacturer.
206	Instruments not free from dirt, oil, polishing compound or foreign matter.
Minor B:	
251	Dimensions (other than 151) not as specified.

4.3.1 Examination of preparation for delivery. An examination shall be made to determine compliance with the packaging, packing and marking requirements of section 5. Defects shall be scored as specified in table IV. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with AQL of 4.0 expressed in terms of percent defective.

TABLE IV. Examination of preparation for delivery

Examine	Defect
Contents	Not as specified.
Container	Not as specified.
Marking	Omitted; incorrect; illegible; improper size, location, sequence or method of application.
Material	Component missing or damaged.
Workmanship	Bulging or distortion of containers.

4.3.1.1 Examination of closure, waterproofing and banding of container. When the shipping container is required to comply with PPP-B-636, the examination for defects in closure, waterproofing or banding shall be in accordance with the appendix of the box specification.

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4.4 Tests. Tests shall be conducted as necessary to determine compliance with specification requirements.

4.4.1 Hardness test. Hardness shall be determined in accordance with ASTM Standard E 18 (Rockwell) or E 92 (Diamond Pyramid Hardness).

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Unit package. Each spatula shall be thoroughly cleaned and dried in accordance with the applicable method in MIL-P-116. Each spatula shall be wrapped in neutral tissue or other noncorrosive material and packaged in accordance with MIL-P-116, method IC-1. A preservative shall not be applied to the spatula.

5.1.1.2 Intermediate package. Twelve unit packages shall be packaged in a snug fitting box constructed in accordance with PPP-B-566, PPP-B-636, class domestic or PPP-B-676. Closure of the box shall be adequate to prevent accidental opening.

5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. Twenty-four intermediate packages (288 units) shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601 overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in the appendix of PPP-B-636.

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5.2.2 Level B. Twenty four intermediate packages (288 units) shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Civil agencies. Marking for shipment shall be in accordance with Fed. Std. No. 123.

5.3.2 Military activities.

5.3.2.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contractor shall be shown.

5.3.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number and date of this specification.
b. Type and style required (see 1.2).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
d. Approval of trade name if applicable (see 3.6).

6.2 Standard sample. For access to standard sample, address the procuring activity issuing the invitation for bids or request for proposal (see 3.1.2, 3.5 and 3.7).

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Preparing activity:

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Project No. 6520-1406

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